hoopR :: CHEAT SHEET

Men's basketball data in R: NBA, NBA G-League, men's college basketball and 23 international league families plus EuroLeague / EuroCup. Tidy play-by-play, box scores, schedules and rosters from ESPN, a full `stats.nba.com` wrapper, KenPom / Torvik / CollegeBasketballData scrapers, Basketball-Reference, Fox and RealGM, plus season-level loaders from the SportsDataverse data releases. Women's sibling: **wehoop**; Python mirror: `sportsdataverse.nba` / `.mbb` in **sportsdataverse-py**. Page 1 of 4: install, surfaces, loaders, credentials, errors, helpers.

v3.2.0

R >= 4.1.0 · httr2 · data.table
hoopR.sportsdataverse.org

GET STARTED

```
install.packages("hoopR")              # CRAN
pak::pak("sportsdataverse/hoopR")      # main
remotes::install_github(
  "sportsdataverse/hoopR")
library(hoopR)
```

1,316 exports (667 in the international families). Every function returns a tibble / data.frame (or a named list of tibbles for multi-table endpoints). Prefix tells you the source; page tells you where to look.

SURFACES BY PREFIX

Prefix	Source	Page
`load_nba_*` `load_mbb_*` `load_nba_stats_*` `load_ncaa_mbb_*`	season parquet/rds releases (zero scraping)	2 · 3
`espn_nba_*` `espn_mbb_*`	ESPN Site / Core APIs, live	2 · 3
`nba_*`	stats.nba.com Stats API (~160 endpoints)	4
`nbagl_*`	G-League (Stats API, league 20)	3
`nba_l2m*` `nba_referee_assignments`	official.nba.com L2M + referee crews	3
`euroleague_*`	EuroLeague / EuroCup APIs (keyless)	3
`fiba_*` `genius_*` `atrium_*` `acb_*` `kbl_*` `...`	23 international families + 148 competition aliases	3
`kp_*`	KenPom (subscription login)	2
`cbbd_*`	CollegeBasketballData API (key)	2
`torvik_*`	Bart Torvik T-Rank	2
`ncaa_mbb_*`	NCAA.com NET + teams	2
`fox_nba_*` `fox_mbb_*`	Fox Sports (Bifrost)	2 · 3
`bref_*`	Basketball-Reference	3
`realgm_*`	RealGM (needs chromote + Chrome)	3
`spotrac_` `hoopshype_` `nbadraft_` `rotowire_`	salaries · mock draft · injuries	3
`*_crosswalk`	ESPN ⇄ NBA Stats / KenPom / Torvik / Fox ids	2 · 3

DATA LOADERS (FAST PATH)

Full seasons from the `sportsdataverse/sportsdataverse-data` GitHub releases (ESPN-derived, tidy, ready to join). Seasons are the **end year** (2024 = 2023-24).

```
pbp  <- load_nba_pbp(seasons = 2022:2024)
tb   <- load_nba_team_box(
          seasons = most_recent_nba_season())
pb   <- load_mbb_player_box(seasons = 2024)
sch  <- load_mbb_schedule(seasons = 2020:2024)
# stream straight into a DB table
load_nba_pbp(2024, dbConnection = con,
             tablename = "nba_pbp")
```

Dataset	`load_nba_*`	`load_mbb_*`
play-by-play	`pbp`	`pbp`
box scores	`team_box` `player_box`	`team_box` `player_box`
schedules	`schedule`	`schedule`
standings	`standings`	`standings`
rosters	`rosters` `game_rosters`	`rosters` `game_rosters`
season stats	`player_stats` `team_stats`	`player_stats` `team_stats`
people	`player_core` `officials` `draft`	`player_core` `officials`
crosswalks	`team_` `schedule_` `player_crosswalk`	same three
shots / value	`shots` `player_impact`	`shots` `ratings` `player_value`
conferences	`groups` `group_seasons` `group_aliases` `team_group_seasons`	same four

Also `load_nba_stats_*` — 18 stats.nba.com-derived release loaders (schedule pbp possessions lineups shots rosters boxscores game_logs season_stats leaguedash standings officials coaches draft) + `update_nba_stats_db()` — and `load_ncaa_mbb_*` — 13 stats.ncaa.org loaders incl. `rapm` / `rapm_within_team`. Group loaders keep each season's own conference alignment.

Local play-by-play database

```
update_nba_db(dbdir = ".",
              dbname = "hoopR_db",
              tblname = "hoopR_nba_pbp")
update_mbb_db()    # RSQLite / DBI (Suggests)
con <- DBI::dbConnect(RSQLite::SQLite(),
                      "hoopR_db")
```

LIVE ESPN + STATS API

```
# ESPN: game_id from espn_*_scoreboard
#       or from load_*_schedule
espn_nba_scoreboard(20230423)   # YYYYMMDD
espn_nba_pbp(game_id = 401071880)
espn_mbb_game_all(401479672)    # pbp + boxes
# stats.nba.com: 10-char game ids, "2023-24"
nba_pbp("0022200021", on_court = TRUE)
nba_leaguedashplayerstats(
  season = year_to_season(2023))
```

ESPN wrappers take `game_id`, `team_id`, `athlete_id`, `season` / `year`. Stats API wrappers mirror the endpoint's query params (snake_case) and return a **named list** of result sets.

CREDENTIALS & OPTIONS

```
# KenPom (active subscription)  ?kp_user_pw
Sys.setenv(KP_USER = "you@example.com",
           KP_PW = "...")
has_kp_user_and_pw()   # TRUE before kp_*()
# CollegeBasketballData  ?register_cbbd
Sys.setenv(CBBD_API_KEY = "XXXX-YOUR-KEY")
has_cbbd_key(); cbbd_key()
# or persist both in ~/.Renviron
usethis::edit_r_environ()
# session-wide proxy (stats.nba.com / ESPN)
options(hoopR.proxy = "http://host:port")
options(hoopR.fox_data_key = "...")  # Fox
```

Stats API requests go through `request_with_proxy()`, which sets the required `x-nba-stats-*` / Referer headers; `hoopR.proxy`, `http_proxy` env or an explicit `proxy=` arg are honoured. `progressively()` + `progressr` give progress bars on multi-season loads.

ERRORS & DEFAULTS

```
r <- tryCatch(acb_pbp(match_id = 105401),
  hoopR_no_data = function(e) NULL)
# 404, nothing there  -> hoopR_no_data
# request failed      -> hoopR_asset_fetch
# 400 / 422           -> hoopR_bad_request
```

International, EuroLeague and officiating wrappers raise classed conditions (all inherit `hoopR_error`; EuroLeague and officiating use `hoopR_fetch_error`, EuroLeague `hoopR_invalid_request`) instead of returning an empty frame. Stats API `season` now defaults to the previous season, `year_to_season(most_recent_nba_season() - 2)` : pass it for the current one.

HELPERS

`most_recent_nba_season()`	current NBA season end year
`most_recent_mbb_season()`	current college season end year
`year_to_season(2023)`	`"2023-24"` for Stats API
`csv_from_url(...)`	fast csv reader behind loaders
`progressively(f, p)`	wrap `f` with a progressor
`login()`	KenPom session (reads `KP_USER` / `KP_PW`)
`hoopR:::request_with_proxy()`	httr2 GET w/ NBA headers + proxy (internal)
`hoopR:::rds_from_url()` `:::check_status()`	internal readers / status guard
`teams_links`	bundled KenPom team + conference names by year (2002-26)
`nba_player_dict()` `nba_team_dict()`	Stats-API id dictionaries
`nba_player_headshot_url()` `nba_team_logo_url()`	CDN media URLs

OVERVIEW TOPICS (?NAME)

`?hoopR_loaders` · `?espn_basketball` · `?nba_stats` · `?kp` · `?register_cbbd` · `?espn_basketball_player_core`. pkgdown reference groups follow the same split.

WIDER SPORTSDATAVERSE

wehoop	WNBA + women's college, same API shape
bigballR	stats.ncaa.org MBB lineups / on-off
cfbfastR · **baseballr** · **fastRhockey**	sister sport packages
sportsdataverse-py	`sportsdataverse.nba` (`nba_stats_*`, `espn_nba_*`, `load_nba_*`) · `sportsdataverse.mbb` (`espn_mbb_*`, `ncaa_mbb_*`)
nba_api	Python stats.nba.com reference
data repos	`sportsdataverse-data` (loaders) · `hoopR-nba-data` · `hoopR-mbb-data` · `hoopR-nba-stats-data`

Men's college basketball :: loaders · ESPN · KenPom · CBD · Torvik

Women's mirror: **wehoop** (`espn_wbb_*`, `load_wbb_*`) · stats.ncaa.org lineups: **bigballR** · Python: `sportsdataverse.mbb`

LOADERS + ESPN · ESPN_MBB_*

SEASON LOADERS

```
library(hoopR)
pbp <- load_mbb_pbp(seasons = 2022:2024)
tb  <- load_mbb_team_box(seasons = 2024)
pb  <- load_mbb_player_box(seasons = 2024)
sch <- load_mbb_schedule(seasons = 2006:2024)
load_mbb_standings(); load_mbb_rosters()
load_mbb_player_stats(); load_mbb_team_stats()
load_mbb_ratings(); load_mbb_shots()
load_mbb_player_value()
load_ncaa_mbb_{pbp,possessions,lineups,shots,
  rapm,team_box,player_box,rosters,...}()
# ^ stats.ncaa.org releases
update_mbb_db(dbdir = ".",
              tblname = "hoopR_mbb_pbp")
```

Releases: `espn_mens_college_basketball_*` tags on `sportsdataverse-data`; MBB pbp runs 2006+.

GAMES (LIVE ESPN)

```
espn_mbb_scoreboard("20221117")   # day slate
espn_mbb_pbp(game_id = 401479672)
espn_mbb_team_box(401479672)
espn_mbb_player_box(401479672)
espn_mbb_game_all(401479672)   # pbp + boxes
espn_mbb_wp(401256760)   # win prob per play
espn_mbb_betting(401256760)
espn_mbb_game_rosters(401256760)
```

Core-API per-game extras (event id): `espn_mbb_game_odds` `_game_predictor` `_game_probabilities` `_game_propbets` `_game_officials` `_game_broadcasts` `_game_situation` `_game_play` `_game_play_personnel` `_game_team_statistics` `_game_team_leaders` `_game_team_linescores` `_game_team_score` `_game_team_roster` `_game_powerindex`.

TEAMS, PLAYERS, LEAGUE

```
espn_mbb_teams(); espn_mbb_conferences()
espn_mbb_rankings(); espn_mbb_standings(2021)
espn_mbb_team_stats(150, year = 2020)
espn_mbb_player_stats(athlete_id = 4593919,
                      year = 2021)
espn_mbb_team_roster(150, season = 2024)
espn_mbb_team_schedule(150, season = 2024)
espn_mbb_player_gamelog(athlete_id, 2024)
```

Also: `espn_mbb_team_current_roster` `_team_record` `_team_injuries` `_team_leaders` `_team_news` `_team_season_statistics` `_player_overview` `_player_splits` `_player_career_stats` `_player_eventlog` `_player_info` `_athletes_index` `_leaders` `_season_leaders` `_injuries` `_news` `_futures` `_powerindex` (BPI) `_tournaments` `_week_rankings` `_season_groups` `_coaches` `_venues` `_calendar`.

KENPOM · KP_* (subscription)

```
Sys.setenv(KP_USER = "you@x.com",
           KP_PW = "***")     # ?kp_user_pw
kp_pomeroy_ratings(2020, 2024)   # years
kp_efficiency(2020, 2024)
kp_fourfactors(2020, 2024)
kp_team_schedule("Florida St.", year = 2022)
kp_team_players(team = "Duke", year = 2024)
kp_playerstats(metric = "eFG", year = 2024)
kp_box(game_id, year)
kp_winprob(game_id, year)
```

Family	Functions
Ratings	`kp_pomeroy_ratings` `kp_pomeroy_archive_ratings` `kp_program_ratings` `kp_trends` `kp_hca` `kp_kpoy`
Team stats	`kp_efficiency` `kp_fourfactors` `kp_teamstats` `kp_pointdist` `kp_height` `kp_foul_trouble` `kp_playerstats`
Team pages	`kp_team_schedule` `kp_team_players` `kp_team_player_stats` `kp_team_history` `kp_team_lineups` `kp_team_depth_chart` `kp_gameplan` `kp_opptracker` `kp_minutes_matrix` `kp_coach_history`
Games	`kp_box` `kp_winprob` `kp_fanmatch(date)` `kp_game_attrs(year, attr)`
Conf / misc	`kp_conf(year, conf)` `kp_confhistory` `kp_confstats` `kp_arenas` `kp_officials` `kp_referee` `kp_player_career`
Auth	`login()` `has_kp_user_and_pw()` `kp_user_email()` `kp_password()`

Team names must match KenPom spelling: `teams_links` lists them, with conferences, by year.

BART TORVIK · TORVIK_*

```
torvik_ratings(year = 2024)   # T-Rank
torvik_team_factors(2024)
torvik_player_stats(2024)
torvik_game_schedule(2024)
torvik_game_stats(2024)
torvik_ncaa_results(2015, 2024, type = "team")
```

COLLEGEBASKETBALLDATA · CBBD_*

```
# key: ?register_cbbd
Sys.setenv(CBBD_API_KEY = "XXXX")
cbbd_games(season = 2024, team = "Duke")
cbbd_plays_game(game_id,
  shooting_plays_only = FALSE)
cbbd_lines(season = 2024, conference = "ACC")
cbbd_ratings_adjusted(season = 2024)
cbbd_stats_player_season(2024, team = "Duke")
```

Family	Functions
Games	`cbbd_games` `cbbd_games_teams` `cbbd_games_players` `cbbd_games_media` `cbbd_scoreboard`
Plays	`cbbd_plays_game` `cbbd_plays_date` `cbbd_plays_team` `cbbd_plays_player` `cbbd_plays_tournament` `cbbd_play_types`
Lineups / subs	`cbbd_lineups_game` `cbbd_lineups_team` `cbbd_substitutions_game` `cbbd_substitutions_team` `cbbd_substitutions_player`
Stats	`cbbd_stats_team_season` `cbbd_stats_team_shooting_season` `cbbd_stats_team_leaderboard` `cbbd_stats_player_season` `cbbd_stats_player_shooting_season`
Ratings / polls	`cbbd_ratings_adjusted` `cbbd_ratings_srs` `cbbd_ratings_elo` `cbbd_rankings`
Lines	`cbbd_lines` `cbbd_lines_providers`
Teams / venues	`cbbd_teams` `cbbd_teams_roster` `cbbd_teams_directory` `cbbd_teams_season_overview` `cbbd_conferences` `cbbd_conferences_history` `cbbd_venues`
Recruiting / draft	`cbbd_recruiting_players` `cbbd_recruiting_teams` `cbbd_recruiting_portal` `cbbd_draft_picks` `cbbd_draft_teams` `cbbd_draft_positions`
Key	`cbbd_key()` `has_cbbd_key()` `check_cbbd_key()`

Most take `season`, `team`, `conference`, `start_date_range` / `end_date_range`, `season_type` filters.

NCAA & FOX

```
ncaa_mbb_NET_rankings()   # current NET
ncaa_mbb_teams(year = 2024, division = 1)
fox_mbb_pbp(game_id)
fox_mbb_boxscore(game_id)
fox_mbb_odds(game_id); fox_mbb_teams()
fox_mbb_teams_all(); fox_mbb_league_leaders()
fox_mbb_team_roster(team_id)
fox_mbb_team_stats(team_id)
fox_mbb_team_gamelog(team_id)
fox_mbb_standings(team_id)
```

Fox ids differ from ESPN ids — bridge with the crosswalks below.

CROSSWALKS (KEYED ON ESPN_TEAM_ID)

```
mbb_team_crosswalk(season = 2024)
# ESPN · KenPom · Torvik · Fox ids per team
mbb_schedule_crosswalk(season = 2024)
mbb_player_crosswalk(season = 2024)
load_mbb_team_crosswalk(2024)   # release
load_mbb_schedule_crosswalk()
load_mbb_player_crosswalk()
```

CONFERENCE HISTORY (GROUPS)

```
load_mbb_team_group_seasons(2002:2025)
load_mbb_groups(); load_mbb_group_seasons()
load_mbb_group_aliases()  # ESPN/NCAA/KP
```

Membership as it was each season, not today's alignment; one `group_id` survives renames.

TYPICAL PIPELINE

```
library(dplyr)
pbp <- load_mbb_pbp(2024)
shots <- pbp |>
  filter(shooting_play,
         !is.na(coordinate_x)) |>
  count(team_id, score_value)
kp <- kp_pomeroy_ratings(2024, 2024)
xw <- load_mbb_team_crosswalk(2024)
tb <- load_mbb_team_box(2024) |>
  left_join(xw,
    by = c("team_id" = "espn_team_id"))
```

Column names above (`shooting_play`, `coordinate_x`, `score_value`) follow the ESPN pbp schema documented in `?load_mbb_pbp`.

ELSEWHERE

Per-game ESPN helpers used by the data repo builders: `helper_espn_mbb_pbp` `_team_box` `_player_box`. Python parity: `sportsdataverse.mbb.espn_mbb_pbp`, `load_mbb_pbp`, `ncaa_mbb_*` (bigballR port).

NBA, G-League & international :: ESPN · EuroLeague · FIBA

LOADERS + ESPN · ESPN_NBA_*

SEASON LOADERS (2002+)

```
library(hoopR)
pbp <- load_nba_pbp(seasons = 2022:2024)
tb  <- load_nba_team_box(2024)
pb  <- load_nba_player_box(2024)
sch <- load_nba_schedule(2002:2024)
load_nba_standings(); load_nba_rosters()
load_nba_player_stats(); load_nba_team_stats()
load_nba_draft(); load_nba_player_core()
update_nba_db(dbdir = ".",
              tblname = "hoopR_nba_pbp")
```

Releases: `espn_nba_*` tags on `sportsdataverse-data`.

GAMES (LIVE ESPN)

```
espn_nba_scoreboard(20230423)   # YYYYMMDD
espn_nba_pbp(game_id = 401071880)
espn_nba_team_box(401071880)
espn_nba_player_box(401071880)
espn_nba_game_all(401283399)    # pbp + boxes
espn_nba_wp(401283399)
espn_nba_betting(401283399)
espn_nba_game_rosters(401283399)
```

Core-API per-event extras: `espn_nba_game_odds` `_game_predictor` `_game_probabilities` `_game_propbets` `_game_officials` `_game_broadcasts` `_game_situation` `_game_play` `_game_play_personnel` `_game_team_statistics` `_game_team_leaders` `_game_team_linescores` `_game_team_records` `_game_powerindex`.

TEAMS, PLAYERS, LEAGUE

```
espn_nba_teams(); espn_nba_standings(2021)
espn_nba_team_stats(team_id = 13, year = 2020)
espn_nba_player_stats(athlete_id = 1966,
                      year = 2022)
espn_nba_team_roster(13)
espn_nba_team_schedule(13, season = 2024)
espn_nba_team_depthchart(team_id = 13)
espn_nba_draft(season = 2024)
espn_nba_transactions(season = 2024)
espn_nba_freeagents(); espn_nba_injuries()
espn_nba_futures()
espn_nba_player_contracts(athlete_id = 1966)
espn_nba_powerindex(season = 2024)   # BPI
```

Also: `espn_nba_athletes_index` `_player_overview` `_player_gamelog` `_player_splits` `_player_career_stats` `_player_awards` `_player_contract` `_team_record` `_team_injuries` `_team_leaders` `_team_news` `_season_draft` `_draft_athletes` `_draft_rounds` `_coaches` `_franchises` `_venues` `_calendar` `_leaders` `_news`.

G-LEAGUE · NBAGL_*

```
nbagl_schedule(season = year_to_season(2023))
nbagl_standings("2023-24")   # league 20
nbagl_players()
nbagl_pbp(game_id, on_court = TRUE)
nbagl_live_pbp(game_id)
nbagl_live_boxscore(game_id)
```

Same Stats-API backend as `nba_*` (page 4); most `nba_*` wrappers also accept `league_id = "20"`.

FOX SPORTS · FOX_NBA_*

```
fox_nba_pbp(game_id)
fox_nba_boxscore(game_id)
fox_nba_odds(game_id); fox_nba_teams()
fox_nba_team_roster(team_id)
fox_nba_team_stats(team_id)
fox_nba_team_gamelog(team_id)
fox_nba_standings(team_id)
fox_nba_league_leaders()
```

CROSSWALKS (KEYED ON ESPN_TEAM_ID)

```
nba_team_crosswalk(season = 2024)
# ESPN · NBA Stats · Fox ids, one row per team
nba_schedule_crosswalk(2024)
nba_player_crosswalk(2024)
load_nba_team_crosswalk(2024)  # release
load_nba_schedule_crosswalk()
load_nba_player_crosswalk()
```

INJURIES, SALARIES, MOCK DRAFTS

```
rotowire_injuries(); bref_injuries()
spotrac_team_cap(season = 2025)   # cap sheets
hoopshype_salaries()              # salaries
nbadraft_mock_draft(year = 2026)  # mock draft
```

OFFICIATING

```
# official.nba.com: L2M reports, ref crews
nba_l2m(game_id = "0042500405")
nba_l2m_games(season = 2025)   # end year
nba_referee_assignments("2025-06-05",
  league = "nba")          # or "gl", "wnba"
```

Typed empty schemas, so `bind_rows()` over many dates works.

EUROLEAGUE · EUROLEAGUE_*

```
euroleague_games("E", "E2025")  # U = EuroCup
euroleague_game_pbp(game_code = 1,
                    season_code = "E2025")
```

15 wrappers: `competitions seasons rounds clubs people games game_stats game_report game_boxscore game_header game_points` (shots) `standings(kind =) player_stats team_stats(mode =)`.

BASKETBALL-REFERENCE · BREF_*

```
bref_players_stats(2025, table = "per_game")
bref_teams_stats(2025, table = "per_game")
bref_player_game_log("jamesle01", 2025)
bref_team_roster(team = "LAL", season = 2025)
bref_standings(2025); bref_draft(2025)
bref_awards(2025); bref_player_bios("a")
```

`table` takes the page's table id (`per_game`, `totals`, `advanced`, ...). Polite scraping: cache results, do not loop aggressively.

REALGM · REALGM_* (chromote + Chrome)

```
realgm_players(); realgm_teams()
realgm_coaches(); realgm_gms()
realgm_player_stats(season = 2025,
  stat_type = "Averages",
  season_type = "Regular_Season")
realgm_team_stats(); realgm_standings()
realgm_transactions(); realgm_salary_cap()
realgm_rookie_scale()
realgm_draft(2025); realgm_draft_prospects()
realgm_early_entry()
realgm_future_free_agents()
realgm_players_abroad()
realgm_individual_games()
realgm_individual_seasons()
```

RealGM sits behind a Cloudflare challenge; the headless-Chrome session is opened lazily on first call.

PER-GAME HELPERS (DATA REPOS)

`helper_espn_nba_pbp` `helper_espn_nba_team_box` `helper_espn_nba_player_box` (and the `_mbb_` trio) wrap the raw ESPN summary into the loader schemas — what `hoopR-nba-data` / `hoopR-mbb-data` run nightly.

INTERNATIONAL · 23 FAMILIES

```
fiba_eurobasket_schedule(season = 2025)
fiba_bcl_pbp(game_id, season = 2025)
acb_pbp(match_id = 105401)    # Liga ACB
kbl_schedule(from_date = "20251101",
             to_date = "20251102")
nbl_rosetta_matches(season = 2026)
genius_competitions(org = "ESAKE")
```

`fiba_` `genius_` `atrium_` `nbl_rosetta_` `scb_` (BBL) `feb_` `tbf_` `aba_` `acb_` `bleague_` `sportpress_` (BNXT) `cba_` `cebl_` `sportality_` (Basketligaen) `easl_` `ibba_` `kbl_` `kzs_` `lba_` `lkl_` `pleague_` `tpbl_` `vtb_`. 148 competition aliases: `fiba_{wc,oly,bcl,bcla,eurobasket,americup,asiacup,afrobasket,icup}_*`, `slb_` `bal_` `nznbl_` `esake_` `lnbm_` `lub_` (Genius), `lnb_` `lnbp_` `nbl1_` `lmb_` `lpb_` (Atrium), each with `_schedule` `_pbp` `_player_box` `_team_box` `_shots`. Names, args and columns match sdv-py; 1 s between requests to a host; proxy via `GENIUS_PROXY` / `CBA_PROXY`.

ANALYTICS UTILITIES (VECTORISED)

```
nba_true_shooting_pct(pts, fga, fta)
nba_effective_fg_pct(fgm, fg3m, fga)
nba_ft_rate(fta, fga)
nba_possessions(fga, fta, oreb, tov)
nba_pace(poss, minutes)
nba_offensive_rating(pts, poss)
nba_net_rating(off_rating, def_rating)
nba_usage_rate(fga, fta, tov, minutes,
  team_fga, team_fta, team_tov, team_minutes)
nba_four_factors(fgm, fg3m, fga, fta, ftm,
  oreb, tov, opp_dreb)
nba_game_score(pts, fgm, fga, fta, ftm, oreb,
  dreb, stl, ast, blk, pf, tov)
box |> nba_add_advanced_metrics()  # adds all
nba_per_minutes(df, cols)
nba_per_possessions(df, cols)
```

Also `nba_defensive_rating` `nba_assist_pct` `nba_assist_to_turnover` `nba_turnover_pct` `nba_oreb_pct`. League-agnostic: works on ESPN or Stats-API box frames.

POSSESSIONS, RAPM, SHOT ZONES

```
poss <- nba_possession_lineups("0022200001")
# off_player_1..5, def_player_1..5, points
rapm <- nba_rapm(poss)   # ridge o/d/total
nba_shot_zones("0022200001")   # FGA by zone
nba_tracking_aggregate(
  seasons = c("2022-23", "2023-24"),
  pt_measure_type = "Drives",
  per_mode = "Totals")
```

RAPM needs a full season of possessions (5-8k) to be meaningful; bind many `nba_possession_lineups()` frames first.

RESULT-SET REDUCERS & TIDY WRAPPERS

```
sets <- nba_playercareerstats("2544")   # list
nba_bind_sets(sets, tag_column = "set_name")
nba_join_sets(sets, join_key = "PLAYER_ID")
nba_nest_sets(sets, keep_cols = "PLAYER_ID")
nba_tidy_boxscore(game_id = "0022200021")
nba_tidy_player_profile(player_id = "2544")
nba_tidy_franchise_history()
```

Media: `nba_player_headshot_url(player_id, size = "1040x760")` · `nba_team_logo_url(team_id, variant = "primary")`.

NBA Stats API :: nba_* wrappers for stats.nba.com

Python siblings: nba_api · sportsdataverse.nba.nba_stats_* · WNBA: wehoop wnba_* · overview ?nba_stats

CONVENTIONS

CALLING PATTERN

```
library(hoopR)
sets <- nba_leaguedashplayerstats(
  season = year_to_season(2023), # "2023-24"
  season_type = "Regular Season",
  per_mode = "PerGame",
  league_id = "00")   # "20" = G-League
names(sets)             # result-set names
df <- sets$LeagueDashPlayerStats  # one tibble
nba_bind_sets(sets)     # or join_/nest_sets
```

Args mirror the endpoint's query string in snake_case; `season` defaults to the previous season. Game ids are 10-char strings (`"0022200021"`); player/team ids are character. Every wrapper returns a **named list** of result sets (V3 boxscores/pbp return nested-JSON-derived frames).

Headers + proxy: all calls go through `request_with_proxy()` ; set `options(hoopR.proxy = ...)` or pass `proxy =` when stats.nba.com blocks your IP (cloud / datacenter ranges often hang). Go slow: `Sys.sleep()` between calls.

PLAY-BY-PLAY & LIVE

`nba_pbp(game_id, on_court = TRUE, version = "v3")`	V2/V3 pbp + on-court lineups
`nba_pbps(game_ids, nest_data = FALSE)`	multi-game pbp
`nba_playbyplayv3` `nba_data_pbp`	raw V3 · data.nba.com feed
`nba_live_pbp` `nba_live_boxscore`	CDN live feeds
`nba_winprobabilitypbp` `nba_gamerotation`	WP overlay · rotations
`nba_todays_scoreboard` `nba_scoreboard` `_scoreboardv2` `_scoreboardv3`	slates
`nba_schedule` `nba_scheduleleaguev2int`	season schedule
`nba_leaguegamefinder` `nba_leaguegamelog`	game finder / log

SHOT CHARTS · SYNERGY · TRACKING

`nba_shotchartdetail` `nba_shotchartleaguewide` `nba_shotchartlineupdetail` `nba_synergyplaytypes` `nba_leaguedashptstats` `nba_leaguedashptdefend` `nba_leaguedashptteamdefend` `nba_leaguedash playerptshot` `nba_leaguedashteamptshot` `nba_leaguedashoppptshot` `nba_shot_zones` `nba_tracking_aggregate` (page 3).

BOX SCORES

V2 (result sets)	V3 (nested)
`nba_boxscore traditionalv2`	`nba_boxscoretraditionalv3`
`nba_boxscore advancedv2`	`nba_boxscoreadvancedv3`
`nba_boxscore fourfactorsv2`	`nba_boxscorefourfactorsv3`
`nba_boxscoremiscv2`	`nba_boxscoremiscv3`
`nba_boxscore scoringv2`	`nba_boxscorescoringv3`
`nba_boxscoreusagev2`	`nba_boxscoreusagev3`
`nba_boxscore playertrackv2`	`nba_boxscoreplayertrackv3`
`nba_boxscore matchups`	`nba_boxscorematchupsv3`
`nba_boxscore summaryv2`	`nba_boxscoresummaryv3`
`nba_boxscore defensive _defensivev2`	`nba_boxscorehustlev2`
`nba_hustlestats boxscore`	`nba_boxscore similarityscore` `nba_glalum boxscoresimilarityscore`

LEAGUE DASHBOARDS

Players	Teams
`nba_leaguedash playerstats`	`nba_leaguedash teamstats`
`nba_leaguedash playerclutch`	`nba_leaguedash teamclutch`
`nba_leaguedash playershotlocations`	`nba_leaguedash teamshotlocations`
`nba_leaguedash playerbiostats`	`nba_leaguedashlineups`
`nba_leaguehustlestats player` `_playerleaders`	`nba_leaguehustlestats team` `_teamleaders`
`nba_league playerondetails`	`nba_leaguelineupviz`
`nba_league seasonmatchups` `nba_matchupsrollup`	`nba_leaguestandings` `_standingsv3` `nba_iststandings`

LEADERS & HOMEPAGE

`nba_leagueleaders` `nba_alltimeleadersgrids` `nba_homepageleaders` `nba_homepagev2` `nba_leaderstiles` `nba_assistleaders` `nba_assisttracker` `nba_dunkscoreleaders` `nba_gravityleaders` `nba_defensehub` `nba_playoffpicture` `nba_commonplayoffseries` `nba_fantasywidget` `nba_infographicfanduelplayer` .

PLAYERS

```
nba_commonallplayers("2023-24")   # id lookup
nba_playerindex(); nba_player_dict()
nba_commonplayerinfo(player_id = "2544")
nba_playercareerstats(player_id = "2544",
                      per_mode = "Totals")
nba_playergamelog("2544", season = "2023-24")
nba_playerprofilev2(player_id = "2544")
```

Dashboards (nba_playerdashboardby*)

`clutch` `gamesplits` `generalsplits` `lastngames` `opponent` `shootingsplits` `teamperformance` `yearoveryear`

Tracking (nba_playerdashpt*)

`pass` `reb` `shotdefend` `shots`

More

`nba_playergamelogs` `nba_playergamestreakfinder` `nba_playernextngames` `nba_playervsplayer` `nba_playercompare` `nba_playerawards` `nba_playerestimatedmetrics` `nba_playerfantasyprofile` `nba_playerfantasyprofilebargraph` `nba_playercareerbycollege` `nba_playercareerbycollegerollup` `nba_playerheadshot` `nba_cumestatsplayer` `nba_cumestatsplayergames`

`nba_tidy_player_profile(player_id)` reduces the profile sets to one frame.

TEAMS

```
nba_teams(); nba_team_dict()
nba_teaminfocommon(team_id = "1610612747")
nba_commonteamroster(team_id, "2023-24")
nba_teamgamelog(team_id, season = "2023-24")
nba_teamyearbyyearstats(team_id)
```

Dashboards (nba_teamdashboardby*)

`clutch` `gamesplits` `generalsplits` `lastngames` `opponent` `shootingsplits` `teamperformance` `yearoveryear`

Tracking / lineups

`nba_teamdashptpass` `nba_teamdashptreb` `nba_teamdashptshots` `nba_teamdashlineups`

More

`nba_teamdetails` `nba_teamestimatedmetrics` `nba_teamgamelogs` `nba_teamgamestreakfinder` `nba_teamhistoricalleaders` `nba_teamplayerdashboard` `nba_teamplayeronoffdetails` `nba_teamplayeronoffsummary` `nba_teamvsplayer` `nba_teamandplayersvsplayers` `nba_commonteamyears` `nba_cumestatsteam` `nba_cumestatsteamgames`

DRAFT & COMBINE

`nba_drafthistory` `nba_draftboard` `nba_draftcombinestats` `nba_draftcombineplayeranthro` `nba_draftcombinedrillresults` `nba_draftcombinespotshooting` `nba_draftcombinenonstationaryshooting` .

FRANCHISES

`nba_franchisehistory` `nba_franchiseplayers` `nba_franchiseleaders` `nba_franchiseleaderswrank` · `nba_tidy_franchise_history()` = active + defunct rows in one frame.

VIDEO

`nba_videodetails` `nba_videodetailsasset` `nba_videoevents` `nba_videoeventsasset` `nba_videostatus` .

COMMON FILTERS

Arg	Typical values
`season`	`"2023-24"` — use `year_to_season(2023)`
`season_type`	`"Regular Season"` `"Playoffs"` `"Pre Season"` `"PlayIn"`
`per_mode`	`"Totals"` `"PerGame"` `"Per100Possessions"` `"Per36"`
`measure_type`	`"Base"` `"Advanced"` `"Misc"` `"Four Factors"` `"Scoring"` `"Opponent"` `"Usage"` `"Defense"`
`league_id`	`"00"` NBA · `"10"` WNBA · `"20"` G-League
`player_or_team`	`"P"` / `"T"` (synergy, game finder)
`date_from date_to`	`"MM/DD/YYYY"` or `""`
`last_n_games month opponent_team_id period`	`0` = all
`start_period end_period`	V3 boxscore / pbp slicing

PIPELINE SKETCH

```
library(dplyr); library(purrr)
ids <- nba_schedule("2023-24")$game_id
pbp <- nba_pbps(game_ids = ids[1:20])
poss <- ids[1:20] |>
  map(nba_possession_lineups) |> bind_rows()
nba_rapm(poss) |> arrange(desc(rapm))
```

Real RAPM needs a whole season — see page 3.

hoopR :: CHEAT SHEET

Men's basketball data in R: NBA, NBA G-League, men's college basketball and 23 international league families plus EuroLeague / EuroCup. Tidy play-by-play, box scores, schedules and rosters from ESPN, a full `stats.nba.com` wrapper, KenPom / Torvik / CollegeBasketballData scrapers, Basketball-Reference, Fox and RealGM, plus season-level loaders from the SportsDataverse data releases. Women's sibling: **wehoop**; Python mirror: `sportsdataverse.nba` / `.mbb` in **sportsdataverse-py**. Page 1 of 4: install, surfaces, loaders, credentials, errors, helpers.

v3.2.0

R >= 4.1.0 · httr2 · data.table
hoopR.sportsdataverse.org

GET STARTED

```
install.packages("hoopR")              # CRAN
pak::pak("sportsdataverse/hoopR")      # main
remotes::install_github(
  "sportsdataverse/hoopR")
library(hoopR)
```

1,316 exports (667 in the international families). Every function returns a tibble / data.frame (or a named list of tibbles for multi-table endpoints). Prefix tells you the source; page tells you where to look.

SURFACES BY PREFIX

Prefix	Source	Page
`load_nba_* load_mbb_* load_nba_stats_* load_ncaa_mbb_*`	season parquet/rds releases (zero scraping)	2 · 3
`espn_nba_* espn_mbb_*`	ESPN Site / Core APIs, live	2 · 3
`nba_*`	stats.nba.com Stats API (~160 endpoints)	4
`nbagl_*`	G-League (Stats API, league 20)	3
`nba_l2m* nba_referee_assignments`	official.nba.com L2M + referee crews	3
`euroleague_*`	EuroLeague / EuroCup APIs (keyless)	3
`fiba_* genius_* atrium_* acb_* kbl_* ...`	23 international families + 148 competition aliases	3
`kp_*`	KenPom (subscription login)	2
`cbbd_*`	CollegeBasketballData API (key)	2
`torvik_*`	Bart Torvik T-Rank	2
`ncaa_mbb_*`	NCAA.com NET + teams	2
`fox_nba_* fox_mbb_*`	Fox Sports (Bifrost)	2 · 3
`bref_*`	Basketball-Reference	3
`realgm_*`	RealGM (needs chromote + Chrome)	3
`spotrac_ hoopshype_ nbadraft_ rotowire_`	salaries · mock draft · injuries	3
`*_crosswalk`	ESPN ⇄ NBA Stats / KenPom / Torvik / Fox ids	2 · 3

DATA LOADERS (FAST PATH)

Full seasons from the `sportsdataverse/sportsdataverse-data` GitHub releases (ESPN-derived, tidy, ready to join). Seasons are the **end year** (2024 = 2023-24).

```
pbp  <- load_nba_pbp(seasons = 2022:2024)
tb   <- load_nba_team_box(
          seasons = most_recent_nba_season())
pb   <- load_mbb_player_box(seasons = 2024)
sch  <- load_mbb_schedule(seasons = 2020:2024)
# stream straight into a DB table
load_nba_pbp(2024, dbConnection = con,
             tablename = "nba_pbp")
```

Dataset	`load_nba_*`	`load_mbb_*`
play-by-play	`pbp`	`pbp`
box scores	`team_box player_box`	`team_box player_box`
schedules	`schedule`	`schedule`
standings	`standings`	`standings`
rosters	`rosters game_rosters`	`rosters game_rosters`
season stats	`player_stats team_stats`	`player_stats team_stats`
people	`player_core officials draft`	`player_core officials`
crosswalks	`team_ schedule_ player_crosswalk`	same three
shots / value	`shots player_impact`	`shots ratings player_value`
conferences	`groups group_seasons group_aliases team_group_seasons`	same four

Also `load_nba_stats_*` — 18 stats.nba.com-derived release loaders (schedule pbp possessions lineups shots rosters boxscores game_logs season_stats leaguedash standings officials coaches draft) + `update_nba_stats_db()` — and `load_ncaa_mbb_*` — 13 stats.ncaa.org loaders incl. `rapm` / `rapm_within_team`. Group loaders keep each season's own conference alignment.

Local play-by-play database

```
update_nba_db(dbdir = ".",
              dbname = "hoopR_db",
              tblname = "hoopR_nba_pbp")
update_mbb_db()   # RSQLite / DBI (Suggests)
con <- DBI::dbConnect(RSQLite::SQLite(),
                      "hoopR_db")
```

LIVE ESPN + STATS API

```
# ESPN: game_id from espn_*_scoreboard
#       or from load_*_schedule
espn_nba_scoreboard(20230423)   # YYYYMMDD
espn_nba_pbp(game_id = 401071880)
espn_mbb_game_all(401479672)    # pbp + boxes
# stats.nba.com: 10-char game ids, "2023-24"
nba_pbp("0022200021", on_court = TRUE)
nba_leaguedashplayerstats(
  season = year_to_season(2023))
```

ESPN wrappers take `game_id`, `team_id`, `athlete_id`, `season` / `year`. Stats API wrappers mirror the endpoint's query params (snake_case) and return a **named list** of result sets.

CREDENTIALS & OPTIONS

```
# KenPom (active subscription)  ?kp_user_pw
Sys.setenv(KP_USER = "you@example.com",
           KP_PW = "...")
has_kp_user_and_pw()   # TRUE before kp_*()
# CollegeBasketballData  ?register_cbbd
Sys.setenv(CBBD_API_KEY = "XXXX-YOUR-KEY")
has_cbbd_key(); cbbd_key()
# or persist both in ~/.Renviron
usethis::edit_r_environ()
# session-wide proxy (stats.nba.com / ESPN)
options(hoopR.proxy = "http://host:port")
options(hoopR.fox_data_key = "...")  # Fox
```

Stats API requests go through `request_with_proxy()`, which sets the required `x-nba-stats-*` / Referer headers; `hoopR.proxy`, `http_proxy` env or an explicit `proxy=` arg are honoured. `progressively()` + `progressr` give progress bars on multi-season loads.

ERRORS & DEFAULTS

```
r <- tryCatch(acb_pbp(match_id = 105401),
  hoopR_no_data = function(e) NULL)
# 404, nothing there  -> hoopR_no_data
# request failed      -> hoopR_asset_fetch
# 400 / 422           -> hoopR_bad_request
```

International, EuroLeague and officiating wrappers raise classed conditions (all inherit `hoopR_error`; EuroLeague and officiating use `hoopR_fetch_error`, EuroLeague `hoopR_invalid_request`) instead of returning an empty frame. Stats API `season` now defaults to the previous season, `year_to_season(most_recent_nba_season() - 2)`: pass it for the current one.

HELPERS

`most_recent_nba_season()`	current NBA season end year
`most_recent_mbb_season()`	current college season end year
`year_to_season(2023)`	`"2023-24"` for Stats API
`csv_from_url(...)`	fast csv reader behind loaders
`progressively(f, p)`	wrap `f` with a progressor
`login()`	KenPom session (reads `KP_USER` / `KP_PW`)
`hoopR::: request_with_proxy()`	httr2 GET w/ NBA headers + proxy (internal)
`hoopR:::rds_from_url()` `:::check_status()`	internal readers / status guard
`teams_links`	bundled KenPom team + conference names by year (2002-26)
`nba_player_dict()` `nba_team_dict()`	Stats-API id dictionaries
`nba_player_headshot_url()` `nba_team_logo_url()`	CDN media URLs

OVERVIEW TOPICS (?NAME)

`?hoopR_loaders` · `?espn_basketball` · `?nba_stats` · `?kp` · `?register_cbbd` · `?espn_basketball_player_core`. pkgdown reference groups follow the same split.

WIDER SPORTSDATAVERSE

wehoop	WNBA + women's college, same API shape
bigballR	stats.ncaa.org MBB lineups / on-off
cfbfastR · baseballr · fastRhockey	sister sport packages
sportsdataverse-py	`sportsdataverse.nba` (`nba_stats_*`, `espn_nba_*`, `load_nba_*`) · `sportsdataverse.mbb` (`espn_mbb_*`, `ncaa_mbb_*`)
nba_api	Python stats.nba.com reference
data repos	`sportsdataverse-data` (loaders) · `hoopR-nba-data` · `hoopR-mbb-data` · `hoopR-nba-stats-data`

Men's college basketball :: loaders · ESPN · KenPom · CBD · Torvik

Women's mirror: **wehoop** (`espn_wbb_*`, `load_wbb_*`) · stats.ncaa.org lineups: **bigballR** · Python: `sportsdataverse.mbb`

LOADERS + ESPN · ESPN_MBB_*

SEASON LOADERS

```
library(hoopR)
pbp <- load_mbb_pbp(seasons = 2022:2024)
tb  <- load_mbb_team_box(seasons = 2024)
pb  <- load_mbb_player_box(seasons = 2024)
sch <- load_mbb_schedule(seasons = 2006:2024)
load_mbb_standings(); load_mbb_rosters()
load_mbb_player_stats(); load_mbb_team_stats()
load_mbb_ratings(); load_mbb_shots()
load_mbb_player_value()
load_ncaa_mbb_{pbp,possessions,lineups,shots,
  rapm,team_box,player_box,rosters,...}()
# ^ stats.ncaa.org releases
update_mbb_db(dbdir = ".",
              tblname = "hoopR_mbb_pbp")
```

Releases: `espn_mens_college_basketball_*` tags on `sportsdataverse-data`; MBB pbp runs 2006+.

GAMES (LIVE ESPN)

```
espn_mbb_scoreboard("20221117")   # day slate
espn_mbb_pbp(game_id = 401479672)
espn_mbb_team_box(401479672)
espn_mbb_player_box(401479672)
espn_mbb_game_all(401479672)   # pbp + boxes
espn_mbb_wp(401256760)   # win prob per play
espn_mbb_betting(401256760)
espn_mbb_game_rosters(401256760)
```

Core-API per-game extras (event id): `espn_mbb_game_odds` `_game_predictor` `_game_probabilities` `_game_propbets` `_game_officials` `_game_broadcasts` `_game_situation` `_game_play` `_game_play_personnel` `_game_team_statistics` `_game_team_leaders` `_game_team_linescores` `_game_team_score` `_game_team_roster` `_game_powerindex`.

TEAMS, PLAYERS, LEAGUE

```
espn_mbb_teams(); espn_mbb_conferences()
espn_mbb_rankings(); espn_mbb_standings(2021)
espn_mbb_team_stats(150, year = 2020)
espn_mbb_player_stats(athlete_id = 4593919,
                      year = 2021)
espn_mbb_team_roster(150, season = 2024)
espn_mbb_team_schedule(150, season = 2024)
espn_mbb_player_gamelog(athlete_id, 2024)
```

Also: `espn_mbb_team_current_roster` `_team_record` `_team_injuries` `_team_leaders` `_team_news` `_team_season_statistics` `_player_overview` `_player_splits` `_player_career_stats` `_player_eventlog` `_player_info` `_athletes_index` `_leaders` `_season_leaders` `_injuries` `_news` `_futures` `_powerindex` (BPI) `_tournaments` `_week_rankings` `_season_groups` `_coaches` `_venues` `_calendar`.

KENPOM · KP_* (subscription)

```
Sys.setenv(KP_USER = "you@x.com",
           KP_PW = "***")     # ?kp_user_pw
kp_pomeroy_ratings(2020, 2024)   # years
kp_efficiency(2020, 2024)
kp_fourfactors(2020, 2024)
kp_team_schedule("Florida St.", year = 2022)
kp_team_players(team = "Duke", year = 2024)
kp_playerstats(metric = "eFG", year = 2024)
kp_box(game_id, year)
kp_winprob(game_id, year)
```

Family	Functions
Ratings	`kp_pomeroy_ratings` `kp_pomeroy_archive_ratings` `kp_program_ratings` `kp_trends` `kp_hca` `kp_kpoy`
Team stats	`kp_efficiency` `kp_fourfactors` `kp_teamstats` `kp_pointdist` `kp_height` `kp_foul_trouble` `kp_playerstats`
Team pages	`kp_team_schedule` `kp_team_players` `kp_team_player_stats` `kp_team_history` `kp_team_lineups` `kp_team_depth_chart` `kp_gameplan` `kp_opptracker` `kp_minutes_matrix` `kp_coach_history`
Games	`kp_box` `kp_winprob` `kp_fanmatch(date)` `kp_game_attrs(year, attr)`
Conf / misc	`kp_conf(year, conf)` `kp_confhistory` `kp_confstats` `kp_arenas` `kp_officials` `kp_referee` `kp_player_career`
Auth	`login()` `has_kp_user_and_pw()` `kp_user_email()` `kp_password()`

Team names must match KenPom spelling: `teams_links` lists them, with conferences, by year.

BART TORVIK · TORVIK_*

```
torvik_ratings(year = 2024)   # T-Rank
torvik_team_factors(2024)
torvik_player_stats(2024)
torvik_game_schedule(2024)
torvik_game_stats(2024)
torvik_ncaa_results(2015, 2024, type = "team")
```

COLLEGEBASKETBALLDATA · CBBD_*

```
# key: ?register_cbbd
Sys.setenv(CBBD_API_KEY = "XXXX")
cbbd_games(season = 2024, team = "Duke")
cbbd_plays_game(game_id,
  shooting_plays_only = FALSE)
cbbd_lines(season = 2024, conference = "ACC")
cbbd_ratings_adjusted(season = 2024)
cbbd_stats_player_season(2024, team = "Duke")
```

Family	Functions
Games	`cbbd_games` `cbbd_games_teams` `cbbd_games_players` `cbbd_games_media` `cbbd_scoreboard`
Plays	`cbbd_plays_game` `cbbd_plays_date` `cbbd_plays_team` `cbbd_plays_player` `cbbd_plays_tournament` `cbbd_play_types`
Lineups / subs	`cbbd_lineups_game` `cbbd_lineups_team` `cbbd_substitutions_game` `cbbd_substitutions_team` `cbbd_substitutions_player`
Stats	`cbbd_stats_team_season` `cbbd_stats_team_shooting_season` `cbbd_stats_team_leaderboard` `cbbd_stats_player_season` `cbbd_stats_player_shooting_season`
Ratings / polls	`cbbd_ratings_adjusted` `cbbd_ratings_srs` `cbbd_ratings_elo` `cbbd_rankings`
Lines	`cbbd_lines` `cbbd_lines_providers`
Teams / venues	`cbbd_teams` `cbbd_teams_roster` `cbbd_teams_directory` `cbbd_teams_season_overview` `cbbd_conferences` `cbbd_conferences_history` `cbbd_venues`
Recruiting / draft	`cbbd_recruiting_players` `cbbd_recruiting_teams` `cbbd_recruiting_portal` `cbbd_draft_picks` `cbbd_draft_teams` `cbbd_draft_positions`
Key	`cbbd_key()` `has_cbbd_key()` `check_cbbd_key()`

Most take `season`, `team`, `conference`, `start_date_range` / `end_date_range`, `season_type` filters.

NCAA & FOX

```
ncaa_mbb_NET_rankings()   # current NET
ncaa_mbb_teams(year = 2024, division = 1)
fox_mbb_pbp(game_id)
fox_mbb_boxscore(game_id)
fox_mbb_odds(game_id); fox_mbb_teams()
fox_mbb_teams_all(); fox_mbb_league_leaders()
fox_mbb_team_roster(team_id)
fox_mbb_team_stats(team_id)
fox_mbb_team_gamelog(team_id)
fox_mbb_standings(team_id)
```

Fox ids differ from ESPN ids — bridge with the crosswalks below.

CROSSWALKS (KEYED ON ESPN_TEAM_ID)

```
mbb_team_crosswalk(season = 2024)
# ESPN · KenPom · Torvik · Fox ids per team
mbb_schedule_crosswalk(season = 2024)
mbb_player_crosswalk(season = 2024)
load_mbb_team_crosswalk(2024)   # release
load_mbb_schedule_crosswalk()
load_mbb_player_crosswalk()
```

CONFERENCE HISTORY (GROUPS)

```
load_mbb_team_group_seasons(2002:2025)
load_mbb_groups(); load_mbb_group_seasons()
load_mbb_group_aliases()   # ESPN/NCAA/KP
```

Membership as it was each season, not today's alignment; one `group_id` survives renames.

TYPICAL PIPELINE

```
library(dplyr)
pbp <- load_mbb_pbp(2024)
shots <- pbp |>
  filter(shooting_play,
         !is.na(coordinate_x)) |>
  count(team_id, score_value)
kp <- kp_pomeroy_ratings(2024, 2024)
xw <- load_mbb_team_crosswalk(2024)
tb <- load_mbb_team_box(2024) |>
  left_join(xw,
    by = c("team_id" = "espn_team_id"))
```

Column names above (`shooting_play`, `coordinate_x`, `score_value`) follow the ESPN pbp schema documented in `?load_mbb_pbp`.

ELSEWHERE

Per-game ESPN helpers used by the data repo builders: `helper_espn_mbb_pbp` `_team_box` `_player_box`. Python parity: `sportsdataverse.mbb.espn_mbb_pbp`, `load_mbb_pbp`, `ncaa_mbb_*` (bigballR port).

NBA, G-League & international :: ESPN · EuroLeague · FIBA

stats.nba.com `nba_*` wrappers: page 4 · WNBA mirror: **wehoop** · Python: `sportsdataverse.nba` / `nbagl`

LOADERS + ESPN · ESPN_NBA_*

SEASON LOADERS (2002+)

```
library(hoopR)
pbp <- load_nba_pbp(seasons = 2022:2024)
tb  <- load_nba_team_box(2024)
pb  <- load_nba_player_box(2024)
sch <- load_nba_schedule(2002:2024)
load_nba_standings(); load_nba_rosters()
load_nba_player_stats(); load_nba_team_stats()
load_nba_draft(); load_nba_player_core()
update_nba_db(dbdir = ".",
              tblname = "hoopR_nba_pbp")
```

Releases: `espn_nba_*` tags on `sportsdataverse-data`.

GAMES (LIVE ESPN)

```
espn_nba_scoreboard(20230423)   # YYYYMMDD
espn_nba_pbp(game_id = 401071880)
espn_nba_team_box(401071880)
espn_nba_player_box(401071880)
espn_nba_game_all(401283399)    # pbp + boxes
espn_nba_wp(401283399)
espn_nba_betting(401283399)
espn_nba_game_rosters(401283399)
```

Core-API per-event extras: `espn_nba_game_odds` `_game_predictor` `_game_probabilities` `_game_propbets` `_game_officials` `_game_broadcasts` `_game_situation` `_game_play` `_game_play_personnel` `_game_team_statistics` `_game_team_leaders` `_game_team_linescores` `_game_team_records` `_game_powerindex`.

TEAMS, PLAYERS, LEAGUE

```
espn_nba_teams(); espn_nba_standings(2021)
espn_nba_team_stats(team_id = 13, year = 2020)
espn_nba_player_stats(athlete_id = 1966,
                      year = 2022)
espn_nba_team_roster(13)
espn_nba_team_schedule(13, season = 2024)
espn_nba_team_depthchart(team_id = 13)
espn_nba_draft(season = 2024)
espn_nba_transactions(season = 2024)
espn_nba_freeagents(); espn_nba_injuries()
espn_nba_futures()
espn_nba_player_contracts(athlete_id = 1966)
espn_nba_powerindex(season = 2024)   # BPI
```

Also: `espn_nba_athletes_index` `_player_overview` `_player_gamelog` `_player_splits` `_player_career_stats` `_player_awards` `_player_contract` `_team_record` `_team_injuries` `_team_leaders` `_team_news` `_season_draft` `_draft_athletes` `_draft_rounds` `_coaches` `_franchises` `_venues` `_calendar` `_leaders` `_news`.

G-LEAGUE · NBAGL_*

```
nbagl_schedule(season = year_to_season(2023))
nbagl_standings("2023-24")   # league 20
nbagl_players()
nbagl_pbp(game_id, on_court = TRUE)
nbagl_live_pbp(game_id)
nbagl_live_boxscore(game_id)
```

Same Stats-API backend as `nba_*` (page 4); most `nba_*` wrappers also accept `league_id = "20"`.

FOX SPORTS · FOX_NBA_*

```
fox_nba_pbp(game_id)
fox_nba_boxscore(game_id)
fox_nba_odds(game_id); fox_nba_teams()
fox_nba_team_roster(team_id)
fox_nba_team_stats(team_id)
fox_nba_team_gamelog(team_id)
fox_nba_standings(team_id)
fox_nba_league_leaders()
```

CROSSWALKS (KEYED ON ESPN_TEAM_ID)

```
nba_team_crosswalk(season = 2024)
# ESPN · NBA Stats · Fox ids, one row per team
nba_schedule_crosswalk(2024)
nba_player_crosswalk(2024)
load_nba_team_crosswalk(2024)  # release
load_nba_schedule_crosswalk()
load_nba_player_crosswalk()
```

INJURIES, SALARIES, MOCK DRAFTS

```
rotowire_injuries(); bref_injuries()
spotrac_team_cap(season = 2025)   # cap sheets
hoopshype_salaries()              # salaries
nbadraft_mock_draft(year = 2026)  # mock draft
```

OFFICIATING

```
# official.nba.com: L2M reports, ref crews
nba_l2m(game_id = "0042500405")
nba_l2m_games(season = 2025)   # end year
nba_referee_assignments("2025-06-05",
  league = "nba")          # or "gl", "wnba"
```

Typed empty schemas, so `bind_rows()` over many dates works.

EUROLEAGUE · EUROLEAGUE_*

```
euroleague_games("E", "E2025")  # U = EuroCup
euroleague_game_pbp(game_code = 1,
                    season_code = "E2025")
```

15 wrappers: `competitions` `seasons` `rounds` `clubs` `people` `games` `game_stats` `game_report` `game_boxscore` `game_header` `game_points` (shots) `standings(kind =)` `player_stats` `team_stats(mode =)`.

BASKETBALL-REFERENCE · BREF_*

```
bref_players_stats(2025, table = "per_game")
bref_teams_stats(2025, table = "per_game")
bref_player_game_log("jamesle01", 2025)
bref_team_roster(team = "LAL", season = 2025)
bref_standings(2025); bref_draft(2025)
bref_awards(2025); bref_player_bios("a")
```

`table` takes the page's table id (`per_game`, `totals`, `advanced`, ...). Polite scraping: cache results, do not loop aggressively.

REALGM · REALGM_* (chromote + Chrome)

```
realgm_players(); realgm_teams()
realgm_coaches(); realgm_gms()
realgm_player_stats(season = 2025,
  stat_type = "Averages",
  season_type = "Regular_Season")
realgm_team_stats(); realgm_standings()
realgm_transactions(); realgm_salary_cap()
realgm_rookie_scale()
realgm_draft(2025); realgm_draft_prospects()
realgm_early_entry()
realgm_future_free_agents()
realgm_players_abroad()
realgm_individual_games()
realgm_individual_seasons()
```

RealGM sits behind a Cloudflare challenge; the headless-Chrome session is opened lazily on first call.

PER-GAME HELPERS (DATA REPOS)

`helper_espn_nba_pbp` `helper_espn_nba_team_box` `helper_espn_nba_player_box` (and the `_mbb_` trio) wrap the raw ESPN summary into the loader schemas — what `hoopR-nba-data` / `hoopR-mbb-data` run nightly.

INTERNATIONAL · 23 FAMILIES

```
fiba_eurobasket_schedule(season = 2025)
fiba_bcl_pbp(game_id, season = 2025)
acb_pbp(match_id = 105401)    # Liga ACB
kbl_schedule(from_date = "20251101",
             to_date = "20251102")
nbl_rosetta_matches(season = 2026)
genius_competitions(org = "ESAKE")
```

`fiba_` `genius_` `atrium_` `nbl_rosetta_` `scb_` (BBL) `feb_` `tbf_` `aba_` `acb_` `bleague_` `sportpress_` (BNXT) `cba_` `cebl_` `sportality_` (Basketligaen) `easl_` `ibba_` `kbl_` `kzs_` `lba_` `lkl_` `pleague_` `tpbl_` `vtb_`. 148 competition aliases: `fiba_{wc,oly,bcl,bcla,eurobasket,americup,asiacup,afrobasket,icup}_*`, `slb_` `bal_` `nznbl_` `esake_` `lnbm_` `lub_` (Genius), `lnb_` `lnbp_` `nbl1_` `lmb_` `lpb_` (Atrium), each with `_schedule` `_pbp` `_player_box` `_team_box` `_shots`. Names, args and columns match sdv-py; 1 s between requests to a host; proxy via `GENIUS_PROXY` / `CBA_PROXY`.

ANALYTICS UTILITIES (VECTORISED)

```
nba_true_shooting_pct(pts, fga, fta)
nba_effective_fg_pct(fgm, fg3m, fga)
nba_ft_rate(fta, fga)
nba_possessions(fga, fta, oreb, tov)
nba_pace(poss, minutes)
nba_offensive_rating(pts, poss)
nba_net_rating(off_rating, def_rating)
nba_usage_rate(fga, fta, tov, minutes,
  team_fga, team_fta, team_tov, team_minutes)
nba_four_factors(fgm, fg3m, fga, fta, ftm,
  oreb, tov, opp_dreb)
nba_game_score(pts, fgm, fga, fta, ftm, oreb,
  dreb, stl, ast, blk, pf, tov)
box |> nba_add_advanced_metrics()  # adds all
nba_per_minutes(df, cols)
nba_per_possessions(df, cols)
```

Also `nba_defensive_rating` `nba_assist_pct` `nba_assist_to_turnover` `nba_turnover_pct` `nba_oreb_pct`. League-agnostic: works on ESPN or Stats-API box frames.

POSSESSIONS, RAPM, SHOT ZONES

```
poss <- nba_possession_lineups("0022200001")
# off_player_1..5, def_player_1..5, points
rapm <- nba_rapm(poss)   # ridge o/d/total
nba_shot_zones("0022200001")   # FGA by zone
nba_tracking_aggregate(
  seasons = c("2022-23", "2023-24"),
  pt_measure_type = "Drives",
  per_mode = "Totals")
```

RAPM needs a full season of possessions (5-8k) to be meaningful; bind many `nba_possession_lineups()` frames first.

RESULT-SET REDUCERS & TIDY WRAPPERS

```
sets <- nba_playercareerstats("2544")   # list
nba_bind_sets(sets, tag_column = "set_name")
nba_join_sets(sets, join_key = "PLAYER_ID")
nba_nest_sets(sets, keep_cols = "PLAYER_ID")
nba_tidy_boxscore(game_id = "0022200021")
nba_tidy_player_profile(player_id = "2544")
nba_tidy_franchise_history()
```

Media: `nba_player_headshot_url(player_id, size = "1040x760")` · `nba_team_logo_url(team_id, variant = "primary")`.

NBA Stats API :: nba_* wrappers for stats.nba.com

Python siblings: **nba_api** · `sportsdataverse.nba.nba_stats_*` · WNBA: **wehoop** `wnba_*` · overview `?nba_stats`

CONVENTIONS

CALLING PATTERN

```
library(hoopR)
sets <- nba_leaguedashplayerstats(
  season = year_to_season(2023), # "2023-24"
  season_type = "Regular Season",
  per_mode = "PerGame",
  league_id = "00")   # "20" = G-League
names(sets)              # result-set names
df <- sets$LeagueDashPlayerStats  # one tibble
nba_bind_sets(sets)      # or join_/nest_sets
```

Args mirror the endpoint's query string in snake_case; `season` defaults to the previous season. Game ids are 10-char strings (`"0022200021"`); player/team ids are character. Every wrapper returns a **named list** of result sets (V3 boxscores/pbp return nested-JSON-derived frames).

Headers + proxy: all calls go through `request_with_proxy()` ; set `options(hoopR.proxy = ...)` or pass `proxy =` when stats.nba.com blocks your IP (cloud / datacenter ranges often hang). Go slow: `Sys.sleep()` between calls.

PLAY-BY-PLAY & LIVE

`nba_pbp(game_id, on_court = TRUE, version = "v3")`	V2/V3 pbp + on-court lineups
`nba_pbps(game_ids, nest_data = FALSE)`	multi-game pbp
`nba_playbyplayv3` `nba_data_pbp`	raw V3 · data.nba.com feed
`nba_live_pbp` `nba_live_boxscore`	CDN live feeds
`nba_winprobabilitypbp` `nba_gamerotation`	WP overlay · rotations
`nba_todays_scoreboard` `nba_scoreboard` `_scoreboardv2` `_scoreboardv3`	slates
`nba_schedule` `nba_scheduleleaguev2int`	season schedule
`nba_leaguegamefinder` `nba_leaguegamelog`	game finder / log

SHOT CHARTS · SYNERGY · TRACKING

`nba_shotchartdetail` `nba_shotchartleaguewide` `nba_shotchartlineupdetail` `nba_synergyplaytypes` `nba_leaguedashptstats` `nba_leaguedashptdefend` `nba_leaguedashptteamdefend` `nba_leaguedashplayerptshot` `nba_leaguedashteamptshot` `nba_leaguedashoppptshot` `nba_shot_zones` `nba_tracking_aggregate` (page 3).

BOX SCORES

V2 (result sets)	V3 (nested)
`nba_boxscore traditionalv2`	`nba_boxscoretraditionalv3`
`nba_boxscore advancedv2`	`nba_boxscoreadvancedv3`
`nba_boxscore fourfactorsv2`	`nba_boxscorefourfactorsv3`
`nba_boxscoremiscv2`	`nba_boxscoremiscv3`
`nba_boxscore scoringv2`	`nba_boxscorescoringv3`
`nba_boxscoreusagev2`	`nba_boxscoreusagev3`
`nba_boxscore playertrackv2`	`nba_boxscoreplayertrackv3`
`nba_boxscore matchups`	`nba_boxscorematchupsv3`
`nba_boxscore summaryv2`	`nba_boxscoresummaryv3`
`nba_boxscore defensive _defensivev2`	`nba_boxscorehustlev2`
`nba_hustlestats boxscore`	`nba_boxscore similarityscore` `nba_glalum boxscoresimilarityscore`

LEAGUE DASHBOARDS

Players	Teams
`nba_leaguedash playerstats`	`nba_leaguedash teamstats`
`nba_leaguedash playerclutch`	`nba_leaguedash teamclutch`
`nba_leaguedash playershotlocations`	`nba_leaguedash teamshotlocations`
`nba_leaguedash playerbiostats`	`nba_leaguedashlineups`
`nba_leaguehustlestats player` `_playerleaders`	`nba_leaguehustlestats team` `_teamleaders`
`nba_league playerondetails`	`nba_leaguelineupviz`
`nba_league seasonmatchups` `nba_matchupsrollup`	`nba_leaguestandings` `_standingsv3` `nba_iststandings`

LEADERS & HOMEPAGE

`nba_leagueleaders` `nba_alltimeleadersgrids` `nba_homepageleaders` `nba_homepagev2` `nba_leaderstiles` `nba_assistleaders` `nba_assisttracker` `nba_dunkscoreleaders` `nba_gravityleaders` `nba_defensehub` `nba_playoffpicture` `nba_commonplayoffseries` `nba_fantasywidget` `nba_infographicfanduelplayer` .

PLAYERS

```
nba_commonallplayers("2023-24")   # id lookup
nba_playerindex(); nba_player_dict()
nba_commonplayerinfo(player_id = "2544")
nba_playercareerstats(player_id = "2544",
                      per_mode = "Totals")
nba_playergamelog("2544", season = "2023-24")
nba_playerprofilev2(player_id = "2544")
```

Dashboards (`nba_playerdashboardby*`)

`clutch` `gamesplits` `generalsplits` `lastngames` `opponent` `shootingsplits` `teamperformance` `yearoveryear`

Tracking (`nba_playerdashpt*`)

`pass` `reb` `shotdefend` `shots`

More

`nba_playergamelogs` `nba_playergamestreakfinder` `nba_playernextngames` `nba_playervsplayer` `nba_playercompare` `nba_playerawards` `nba_playerestimatedmetrics` `nba_playerfantasyprofile` `nba_playerfantasyprofilebargraph` `nba_playercareerbycollege` `nba_playercareerbycollegerollup` `nba_playerheadshot` `nba_cumestatsplayer` `nba_cumestatsplayergames`

`nba_tidy_player_profile(player_id)` reduces the profile sets to one frame.

TEAMS

```
nba_teams(); nba_team_dict()
nba_teaminfocommon(team_id = "1610612747")
nba_commonteamroster(team_id, "2023-24")
nba_teamgamelog(team_id, season = "2023-24")
nba_teamyearbyyearstats(team_id)
```

Dashboards (`nba_teamdashboardby*`)

`clutch` `gamesplits` `generalsplits` `lastngames` `opponent` `shootingsplits` `teamperformance` `yearoveryear`

Tracking / lineups

`nba_teamdashptpass` `nba_teamdashptreb` `nba_teamdashptshots` `nba_teamdashlineups`

More

`nba_teamdetails` `nba_teamestimatedmetrics` `nba_teamgamelogs` `nba_teamgamestreakfinder` `nba_teamhistoricalleaders` `nba_teamplayerdashboard` `nba_teamplayeronoffdetails` `nba_teamplayeronoffsummary` `nba_teamvsplayer` `nba_teamandplayersvsplayers` `nba_commonteamyears` `nba_cumestatsteam` `nba_cumestatsteamgames`

DRAFT & COMBINE

`nba_drafthistory` `nba_draftboard` `nba_draftcombinestats` `nba_draftcombineplayeranthro` `nba_draftcombinedrillresults` `nba_draftcombinespotshooting` `nba_draftcombinenonstationaryshooting` .

FRANCHISES

`nba_franchisehistory` `nba_franchiseplayers` `nba_franchiseleaders` `nba_franchiseleaderswrank` · `nba_tidy_franchise_history()` = active + defunct rows in one frame.

VIDEO

`nba_videodetails` `nba_videodetailsasset` `nba_videoevents` `nba_videoeventsasset` `nba_videostatus` .

COMMON FILTERS

Arg	Typical values
`season`	`"2023-24"` — use `year_to_season(2023)`
`season_type`	`"Regular Season"` `"Playoffs"` `"Pre Season"` `"PlayIn"`
`per_mode`	`"Totals"` `"PerGame"` `"Per100Possessions"` `"Per36"`
`measure_type`	`"Base"` `"Advanced"` `"Misc"` `"Four Factors"` `"Scoring"` `"Opponent"` `"Usage"` `"Defense"`
`league_id`	`"00"` NBA · `"10"` WNBA · `"20"` G-League
`player_or_team`	`"P"` / `"T"` (synergy, game finder)
`date_from` `date_to`	`"MM/DD/YYYY"` or `""`
`last_n_games` `month` `opponent_team_id` `period`	`0` = all
`start_period` `end_period`	V3 boxscore / pbp slicing

PIPELINE SKETCH

```
library(dplyr); library(purrr)
ids <- nba_schedule("2023-24")$game_id
pbp <- nba_pbps(game_ids = ids[1:20])
poss <- ids[1:20] |>
  map(nba_possession_lineups) |> bind_rows()
nba_rapm(poss) |> arrange(desc(rapm))
```

Real RAPM needs a whole season — see page 3.